2026 Workshop: Updating urban segregation computation with R

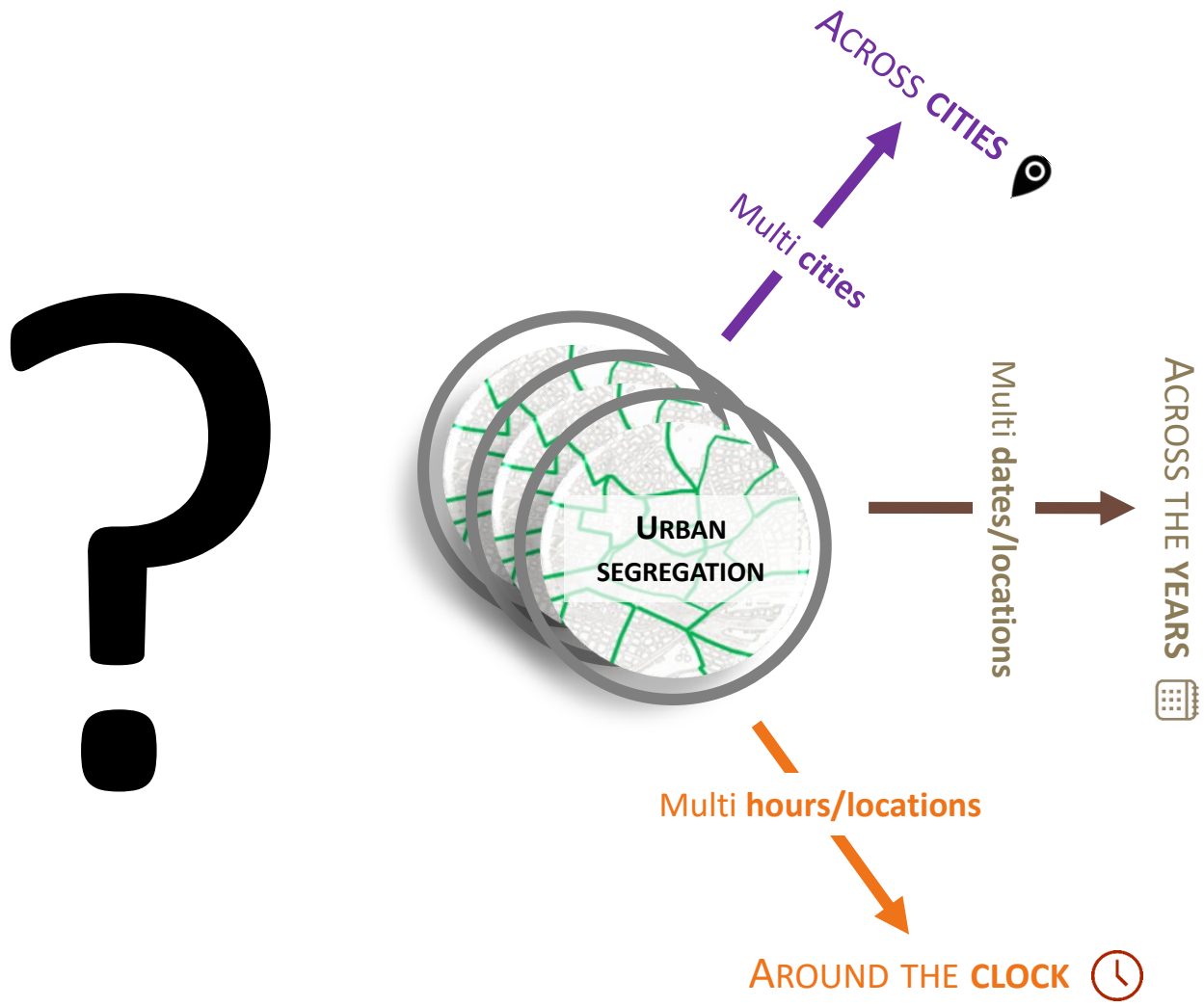
9-13th November 2026

What are the temporal characteristics of urban segregation?

Julie Vallée

CNRS, Toulouse, France

URBAN SEGREGATION OVER **TIME**



KEY POINTS OF THE TALK

I. BRIEF REVIEW OF LITTERATURE

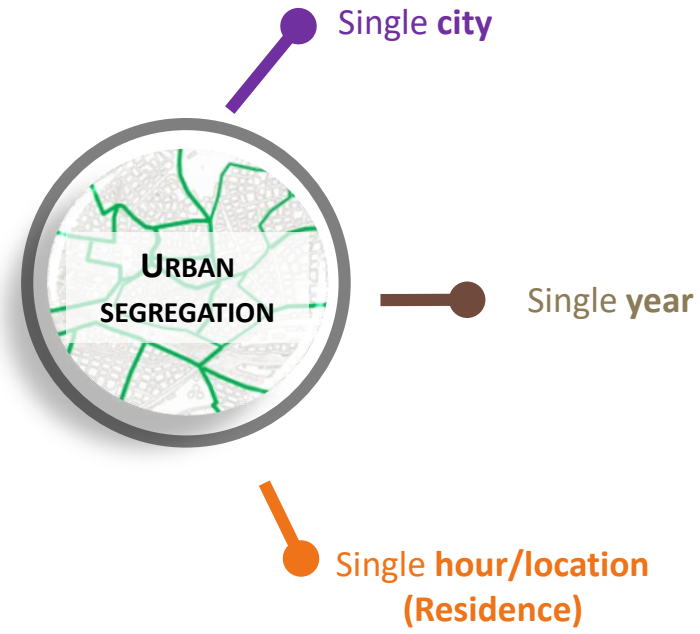
II. METHODOLOGICAL APPROACH

III. EMPIRICAL FINDINGS

VI. PERSPECTIVES

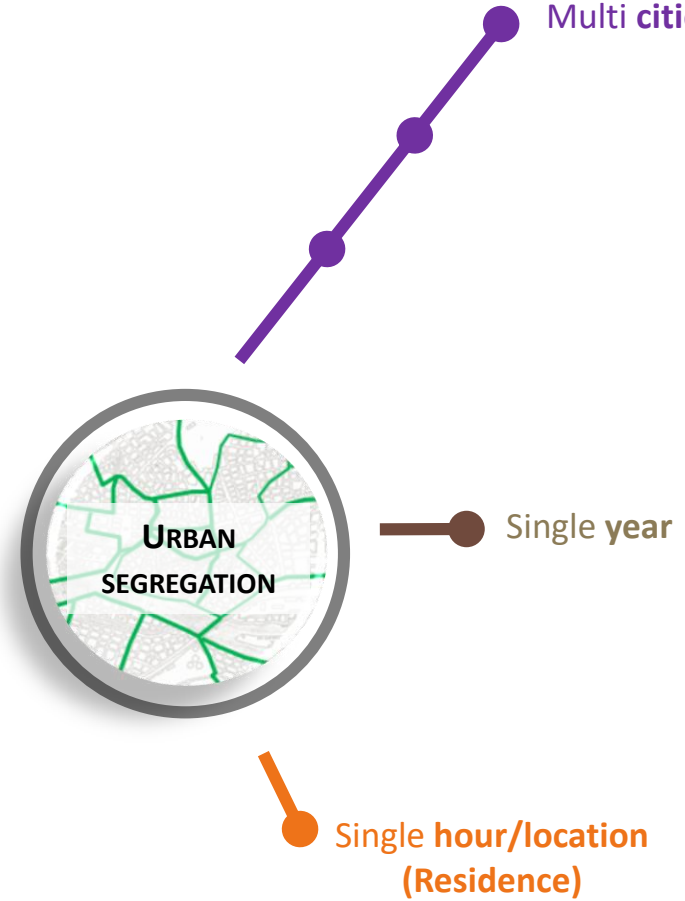
I. Brief review of litterature

URBAN SEGREGATION OVER TIME ?



Many studies
with a single approach !

URBAN SEGREGATION OVER TIME ?



Also many studies comparing segregation across cities with a single temporal approach

Tijdschrift voor economische en sociale geografie — Journal of Economic and Human Geography

Full Access

SEGREGATION AND HOUSING OF MINORITY ETHNIC GROUPS IN WESTERN EUROPEAN CITIES

SAKO MUSTERD, RONALD VAN KEMPEN

First published: 20 October 2009 | https://doi.org/10.1111/j.1467-9663.2009.00558.x | Citations: 58

URBAN GEOGRAPHY
2023, VOL. 44, NO. 1, 1–36
https://doi.org/10.1080/02723638.2021.1959778

Routledge
Taylor & Francis Group

OPEN ACCESS

Socio-economic segregation in European cities. A comparative study of Brussels, Copenhagen, Amsterdam, Oslo and Stockholm

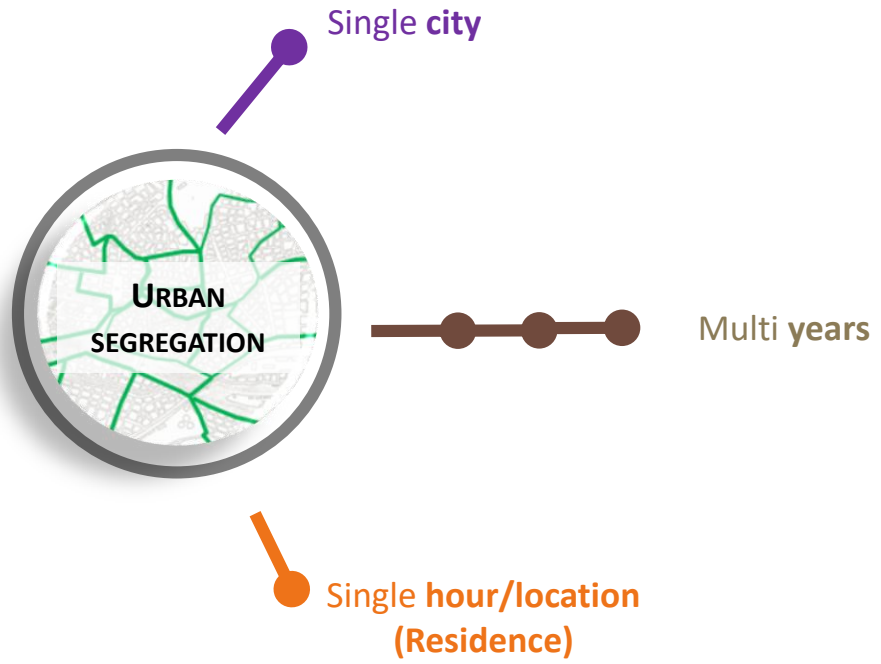
Karen Haandrikman[a], Rafael Costa[b,c], Bo Malmberg[a], Adrian Farner Rogne[d] and Bart Sleutjes[c*]

Journal of Gerontology
1978, Vol. 33, No. 3, 446-453

Residential Segregation by Age in American Metropolitan Areas[1]

Donald O. Cowgill

URBAN SEGREGATION OVER TIME ?



And with a multi-year approach…
but only in one city

EDMOND PRÉTECEILLE

Sociétés Contemporaines (2006) n° 62 (p. 69-93)

LA SÉGRÉGATION SOCIALE A-T-ELLE AUGMENTÉ ?

LA MÉTROPOLE PARISIENNE ENTRE POLARISATION ET MIXITÉ

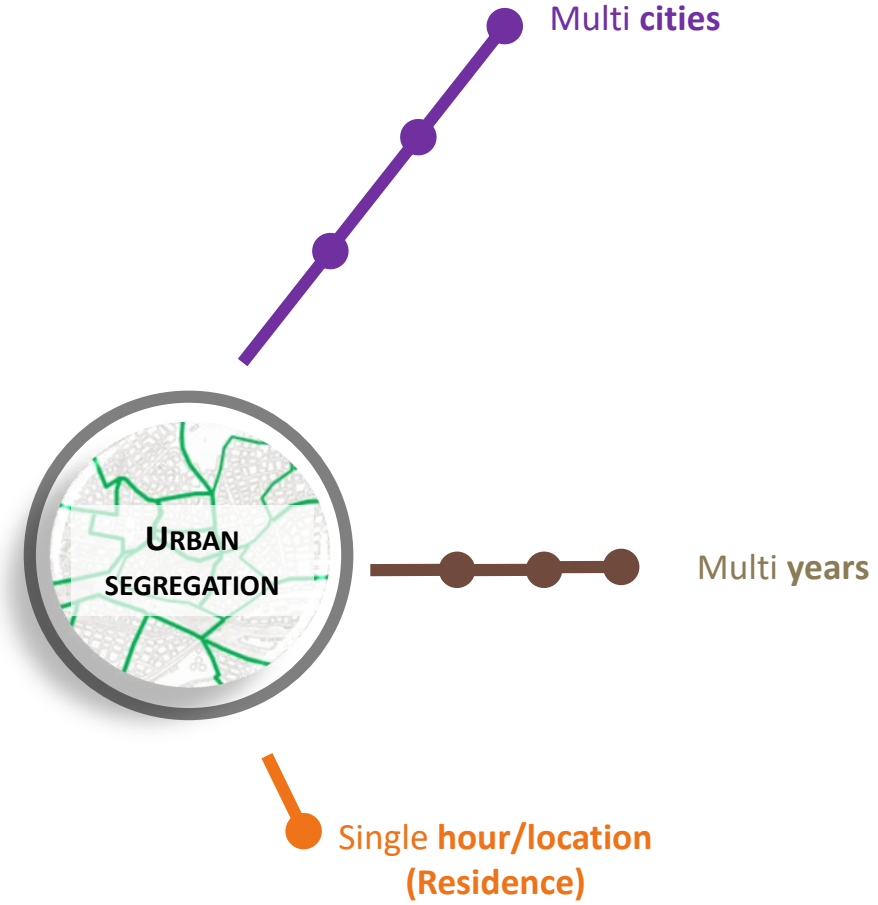
Pour la métropole parisienne, les indices de ségrégation les plus forts en 1982, 1990 et 1999, qu'ils soient calculés selon les IRIS ou les communes et quartiers, sont pour les chefs d'entreprise et les professions libérales ainsi que pour les professions de l'information, des arts et du spectacle.

!! Risky to compare segregation over the years…

- when population distribution changes

- when city boudaries evolves

URBAN SEGREGATION OVER TIME ?



!! Risky to compare segregation over the years…

- when population distribution changes

- when city boudaries evolves

And with a multi-year approach…
for many cities (of the same country)

Urban Studies
45(3) 467–499, March 2008

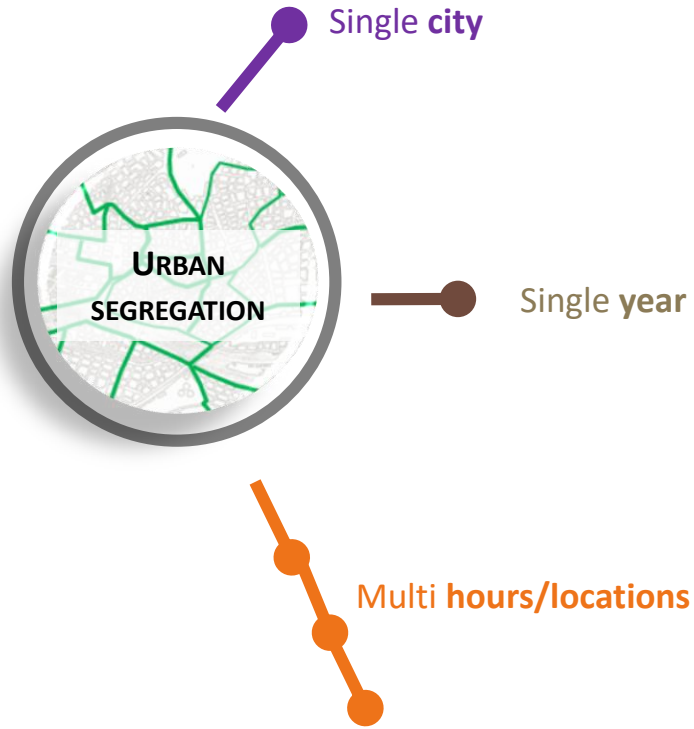
Bifurcation, Fragmentation or Integration? The Racial and Geographical Structure of US Metropolitan Segregation, 1990–2000

Chad R. Farrell

[Paper first received, November 2006; in final form, February 2007]

This research focuses on 97 of the 100 largest US metropolitan areas in the year 2000. It incorporates census data from the summary files of the 1990 and 2000 decennial censuses,

URBAN SEGREGATION OVER TIME ?



Growing number of empirical papers
« beyond residential segregation »
but in a single city & for a single year

Social segregation around the clock in the Paris region (France)

Guillaume Le Roux *, Julie Vallée, Hadrien Commenges

Measuring segregation using patterns of daily travel behavior: A social interaction based model of exposure

Steven Farber [a,*], Morton O'Kelly [b], Harvey J. Miller [b], Tijs Neutens [c]

Beyond residential segregation: A spatiotemporal approach to examining multi-contextual segregation

Yoo Min Park*, Mei-Po Kwan

The temporal variation of ethnic segregation in a city: Evidence from a mobile phone use dataset

Siiri Silm [a,*], Rein Ahas [a,b]

From residence to movement: The nature of racial segregation in everyday urban mobility

Jennifer Candipan
Brown University and Harvard University, USA

Nolan Edward Phillips
Harvard University, USA

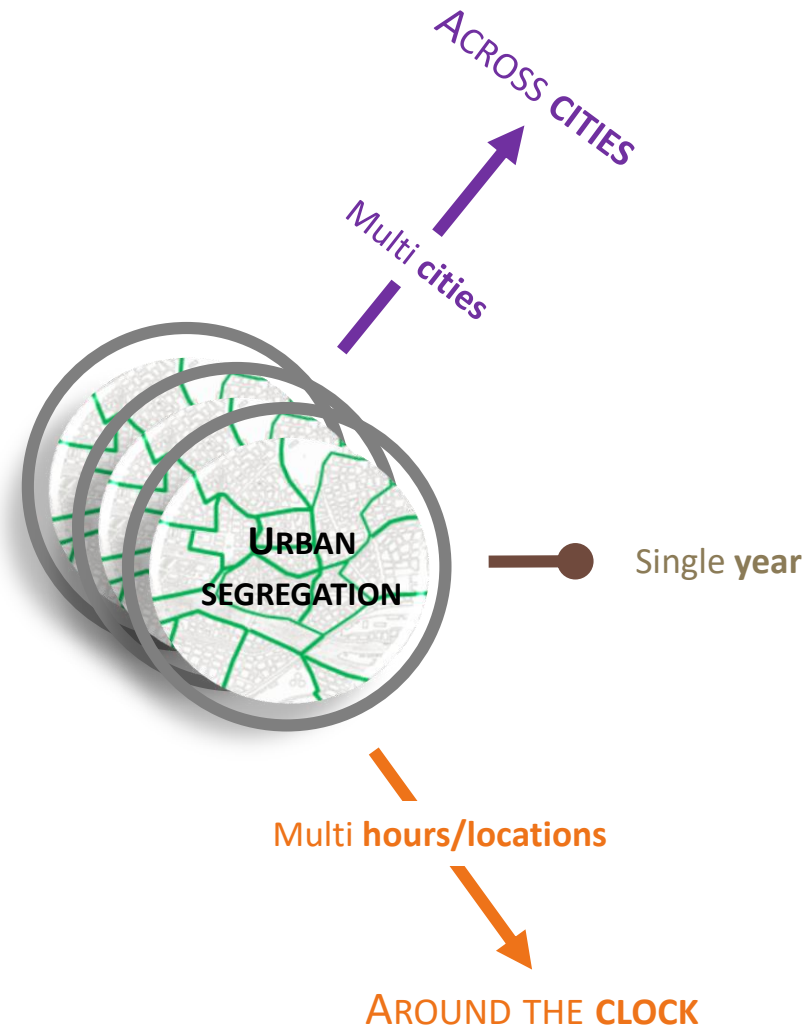
Robert J Sampson
Harvard University, USA

Mario Small
Harvard University, USA

Less risky to compare segregation at different times of the day :

- Stability in population distribution

- Stability in city boudaries



LET'S NOW TAKE A LOOK TOGETHER AT THE DATA AND FINDINGS ON HOURLY VARIATIONS IN SEGREGATION ACROSS CITIES

II. METHODOLOGICAL APPROACH

USE OF TRAVEL SURVEYS FOR EXPLORING HOURLY VARIATIONS IN URBAN SEGREGATION

	Census	Mobile phone	Mobile phone apps with GPS	Ticketing	**Travel survey / OD survey**
Mode of data collection	*Self-reporting*	Passive	Passive	Passive	Self-reporting
Collection frequency	*one-time*	continuous	continuous	continuous	one-time
Locations	*Residential and work places*	All locations (phone on)	All locations (Apps and GPS in active mode)	Origin (and destination) of each trip (in the transport system)	Origin and destination of each trip (on the day of the survey)
Spatial accuracy	*Census tracts*	Cell tower	X; Y	X; Y (stations)	Districts
Temporal accuracy	***No***	hh:mn	hh:mn	hh:mn	hh:mn
Personal information	*Yes*	**No (or very few)**	**No (or very few)**	**No (or very few)**	Yes
Trip information (mode & purpose)	***No***	**No**	**No**	Mode : Yes Purpose : **No**	Yes
Sample size	*Quasi-exhaustive*	Very large	It depends (but no representative)	It depends (but no representative)	Small (but representative)



Data Processing

LIMITS AND STRENGTHS

	Travel survey / OD survey
Mode of data collection	Self-reporting
Data-collection frequency	one-time
Locations	Origin and destination of each trip (on the day of the survey)
Spatial accuracy	Districts
Temporal accuracy	hh:mn
Personal information	Yes
Trip information (mode & purpose)	Yes
Sample size	Small (but representative)

Limitations related to…

>> the **sample size:**

- Risky to cross **multiple socio-demographic indicators**
- Risky to drill down to a **spatial granularity finer than the district**
- Only presence of **residents** in the region

>> the **frequency** of data collection:

- Impossible to study **weekly, monthly, or annual variations**

And strengths:

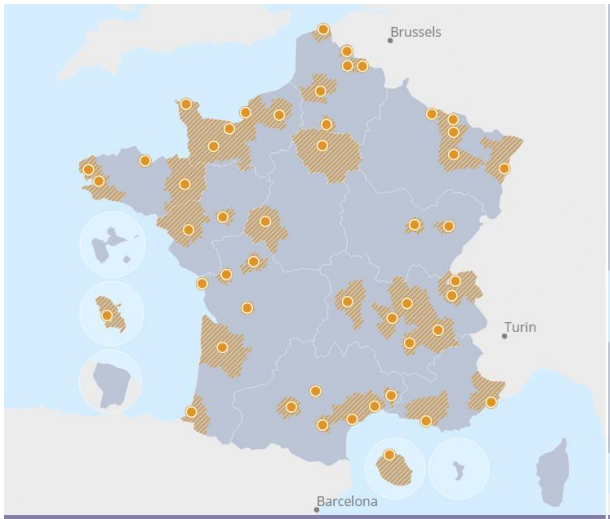
- Information on **purposes, modes of transport**, and **individual socio-demographic profiles** without risky interpolations
- Good **representativeness of the population** in the studied territories
- **Reproducibility & international comparison**

- Initial data : **58** Origin-Destination (O-D) surveys providing a detailed picture of the trip patterns of a city's or region's residents
  - Only respondent's aged 16 years and more

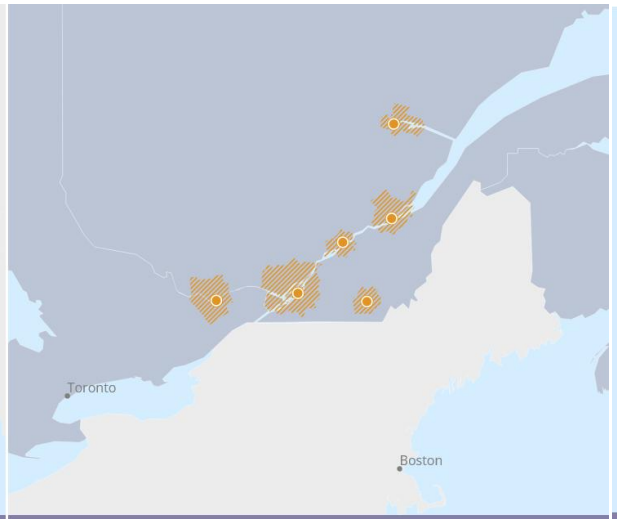
- We transform **trip datasets** into **location datasets** to get 24 hourly time steps of respondents' locations **at exact hours** (04.00, 05.00 etc.).
  - Only trips during the week (Monday-Friday) → hourly location during a **typical weekday**

- For **each hour**, we estimate the **ambient population** per **district** (taking into account weighting coefficients)





IN FRANCE



IN QUEBEC, CANADA

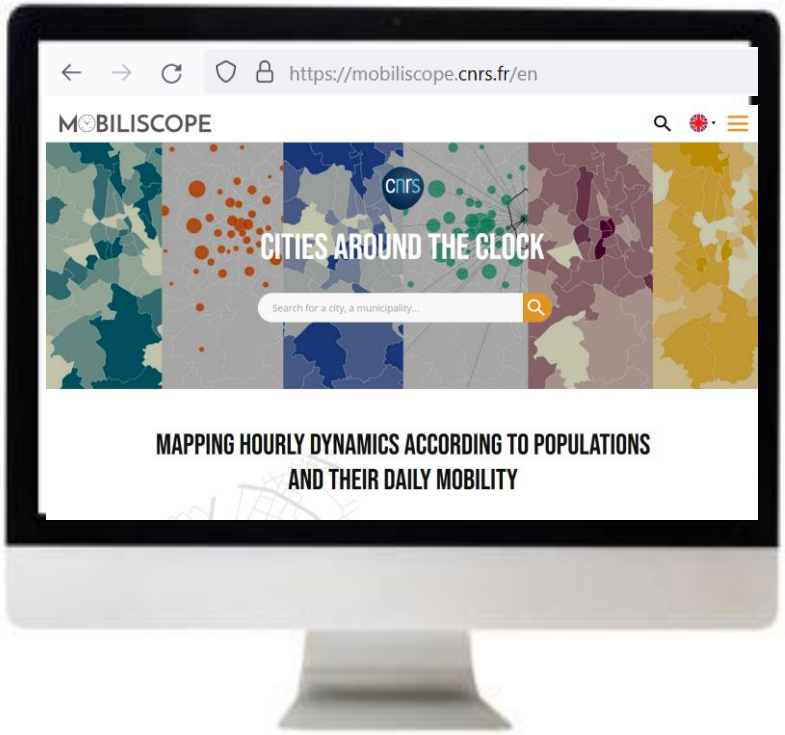


IN LATINE AMERICA

Mobiliscope, an interactive tool available online :

www.mobiliscope.cnrs.fr

Availalable in three languages :
French, English & Spanish



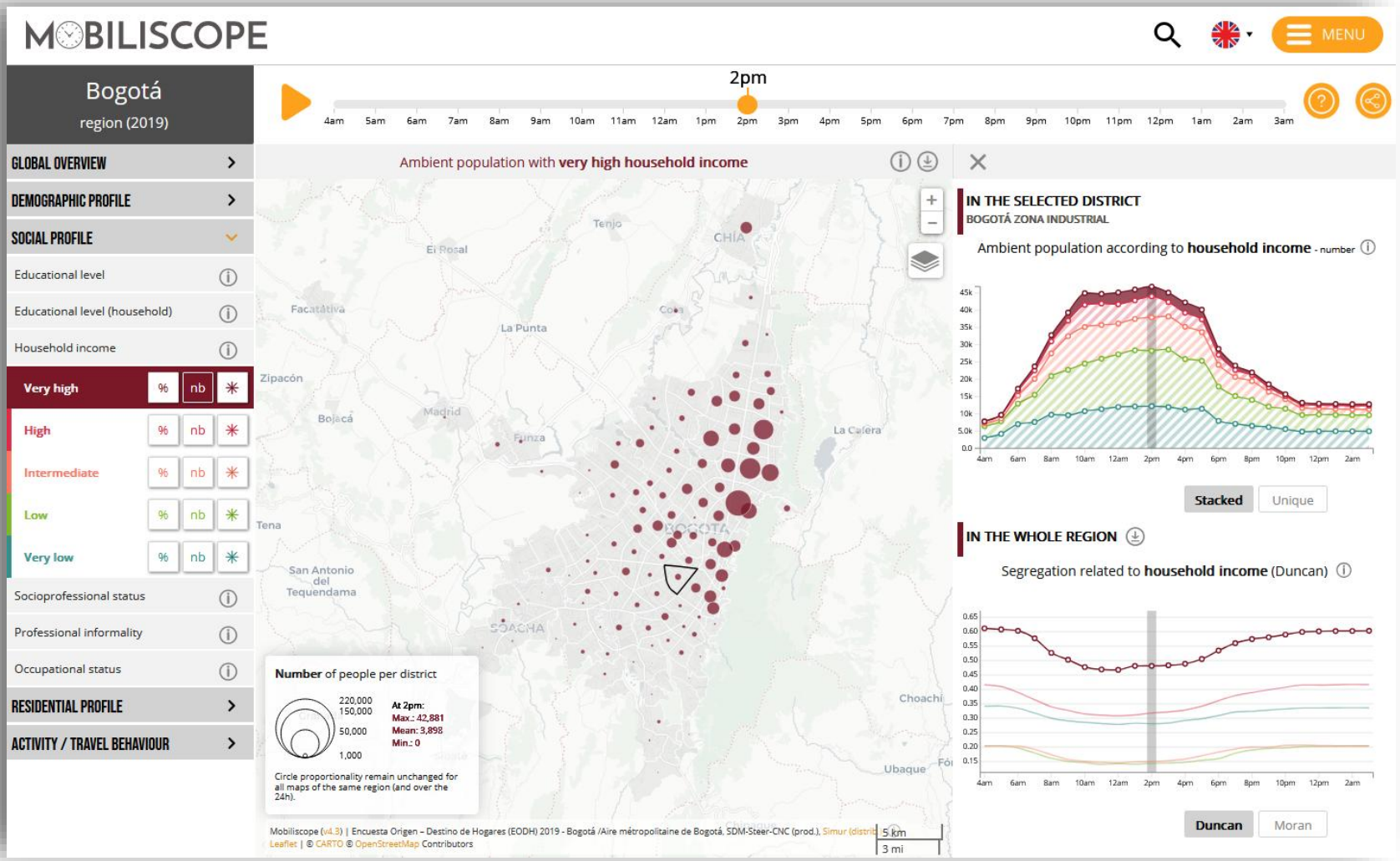
Developed in the spirit of **open science**

- **Programs** → available in 2 public GitLab repositories
  - ✓ One for data processing
  - ✓ One for online plateform

- **Data** → available under open ODbL ense
  - Either in the Mobiliscope - button
  - Or in zenodo (full dataset) **=>** **https://zenodo.org/records/11111161** **(availlable for our workshop!)**

For every city region, it displays **district hourly population** estimations according to :

- people's demographical and social profile (**gender**, **age groups, household composition**, **education** level, **income, occupational** status..)
- their current **activities** and their last **mode of transportation**



TWO MEASURES OF UNIGROUP SEGREGATION

- Dissimilarity index (Duncan)
  - Values express the proportion of individuals of a given group who would have to move to another district (without being replaced) to get similar proportion of people from this group in every district
  - Range from 0 (minimum segregation) to 1 (maximum segregation).

- Spatial autocorrelation index (Moran)
  - To express correlation in social composition among nearby districts.
  - Range from -1 (nearby districts with dissimilar social composition - negative autocorrelation) to 1 (nearby districts with similar social composition - positive autocorrelation). A value close to zero expresses no spatial structure

And why focus on these two measures ?

- *Accessible for non-expert users*
- *Complementary*

TWO COMPLEMENTARY MEASURES

- Dissimilarity index (Duncan)
- Spatial autocorrelation index (Moran)

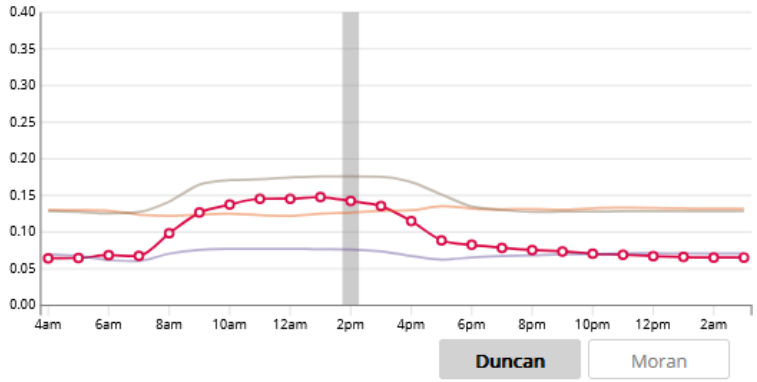
MOBILISCOPE

Montreal
region (2013)

Age groups
- 65 and more
- 35-64
- 25-34
- 15-24

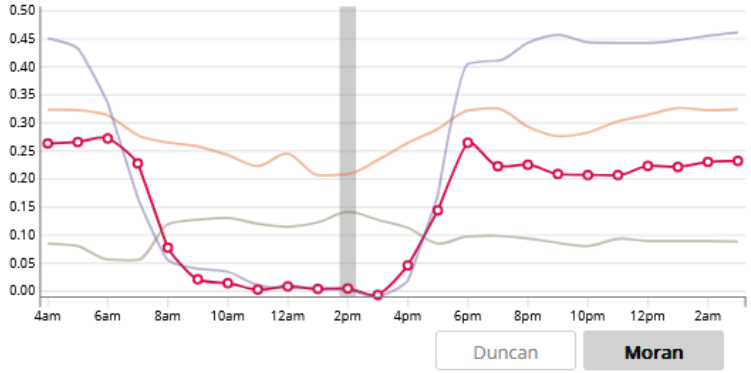
IN THE WHOLE REGION

Segregation related to **age groups** (Duncan)



IN THE WHOLE REGION

Spatial proximity of population with same **age groups** (Moran)



http://mobiliscope.cnrs.fr/en



II. Empirical findings

about hourly variations in segregation across cities

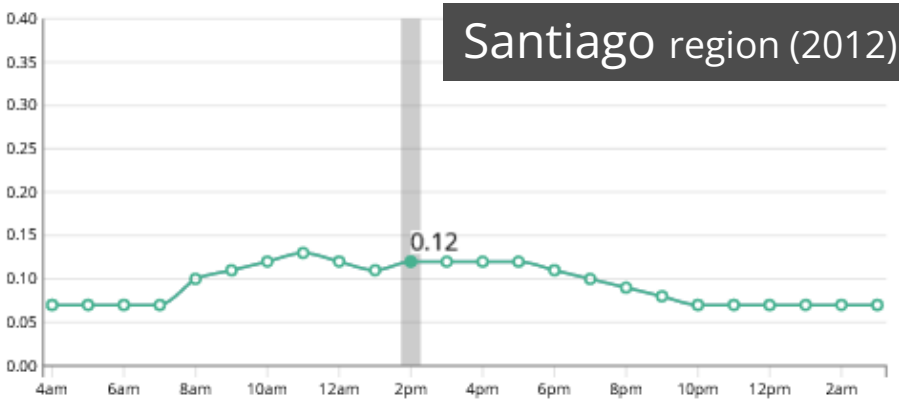
Sex

Female

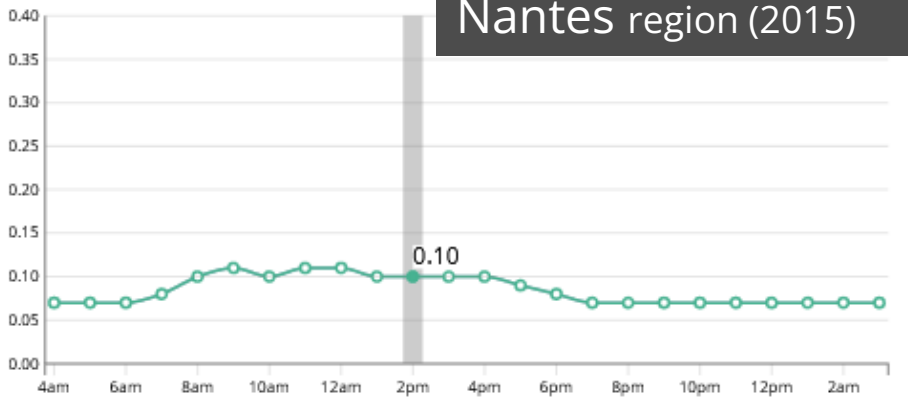
Male

Duncan values in 5 city regions for Women (*vs.* Men)

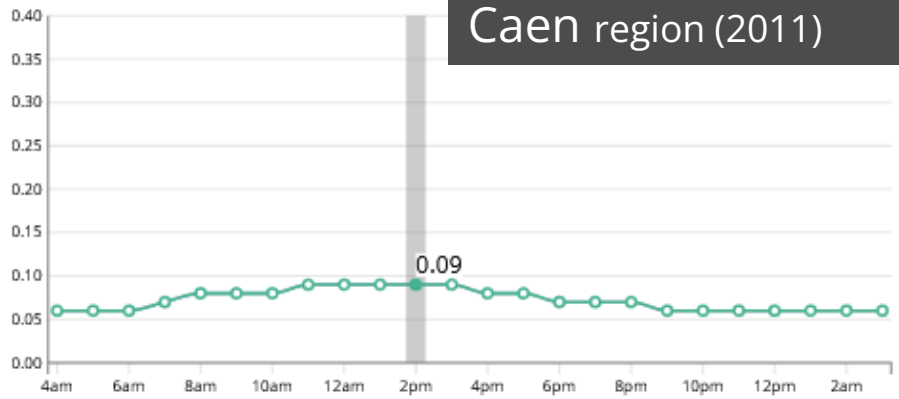
Santiago region (2012)



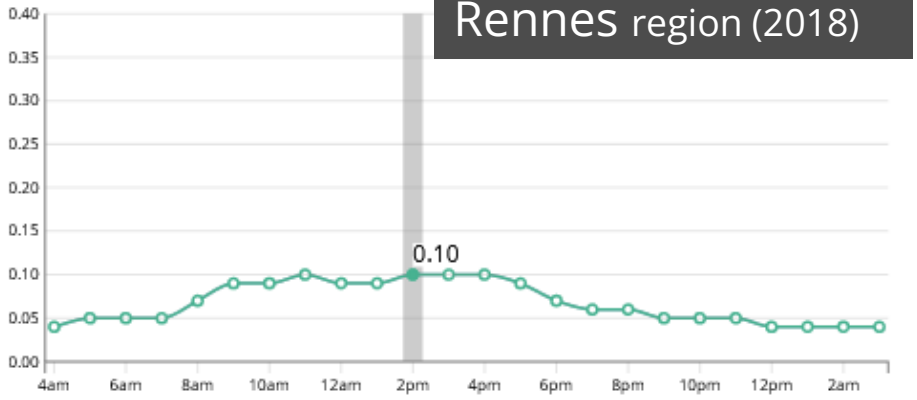
Nantes region (2015)



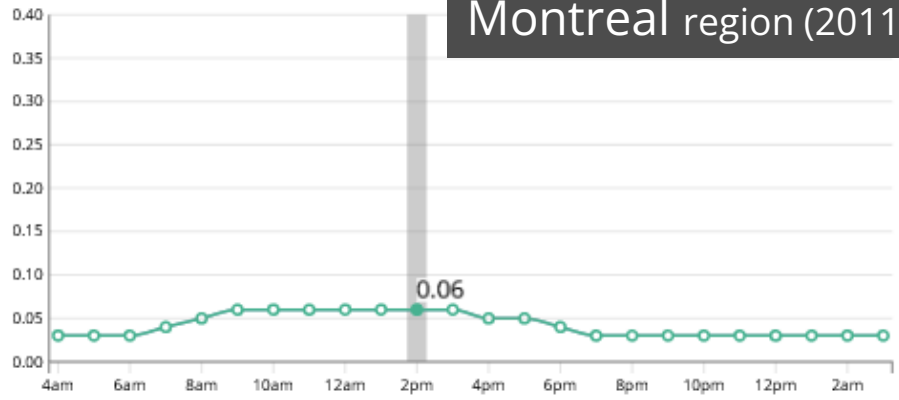
Caen region (2011)



Rennes region (2018)



Montreal region (2011)

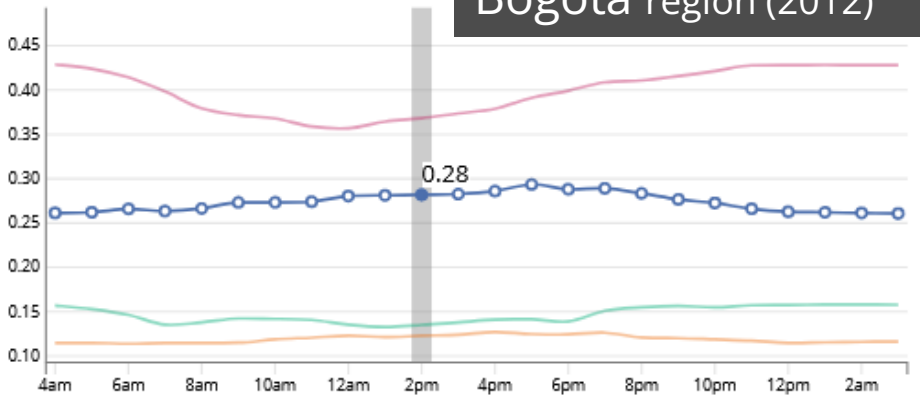


Educational level

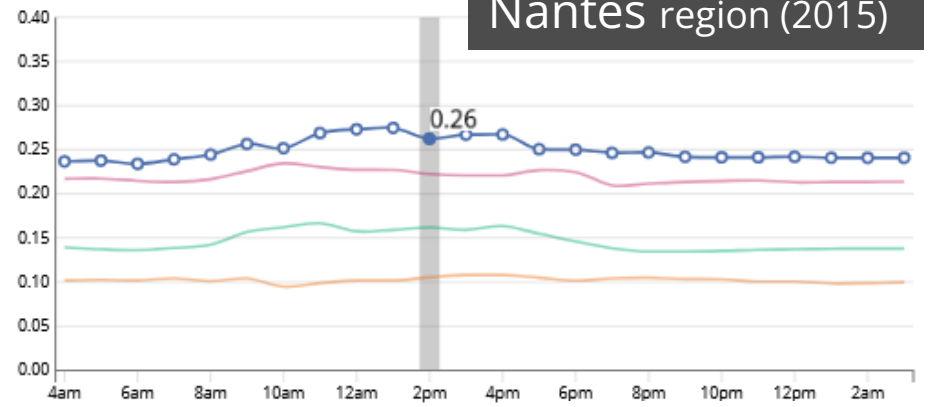
- Very high
- High
- Middle
- Low

DUNCAN VALUES IN 5 CITY REGIONS FOR **LOW EDUC.** SUBPGROUP

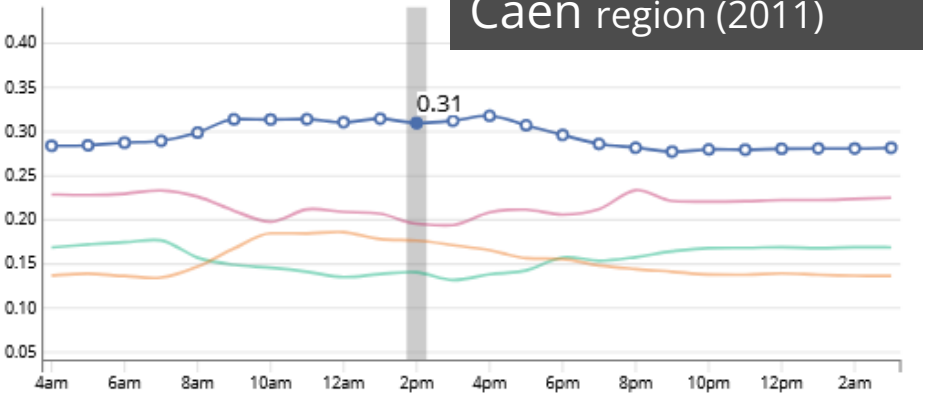
Bogotá region (2012)



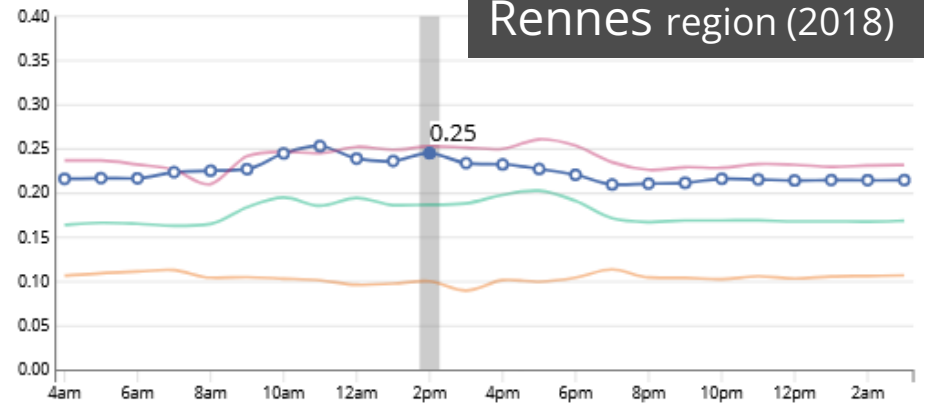
Nantes region (2015)



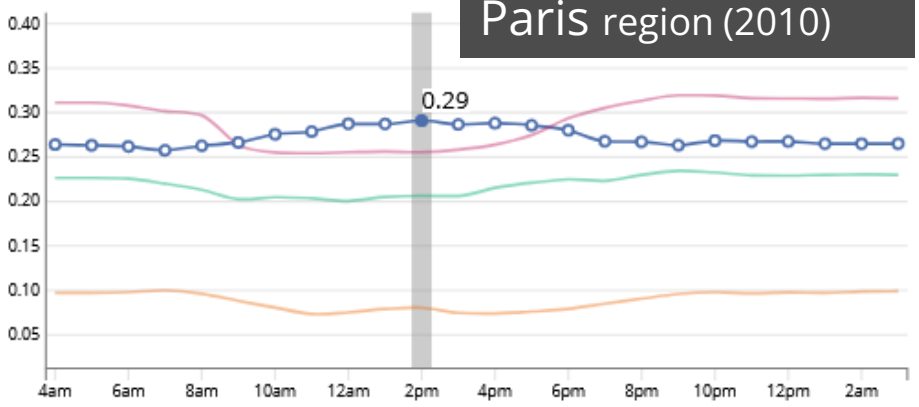
Caen region (2011)



Rennes region (2018)



Paris region (2010)

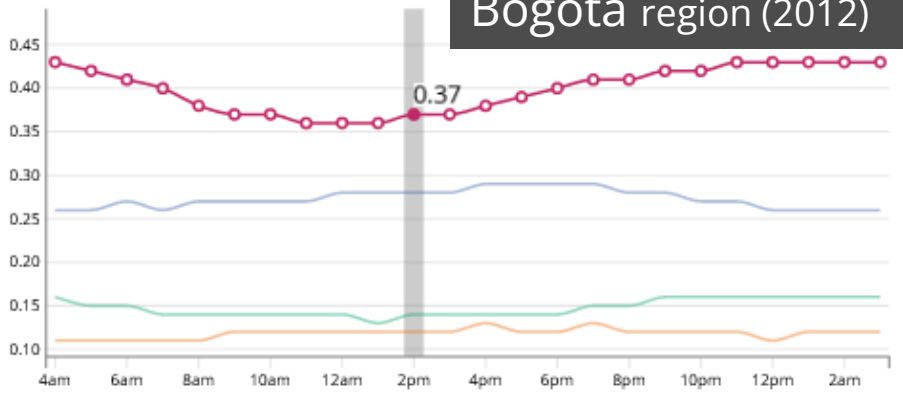


Educational level

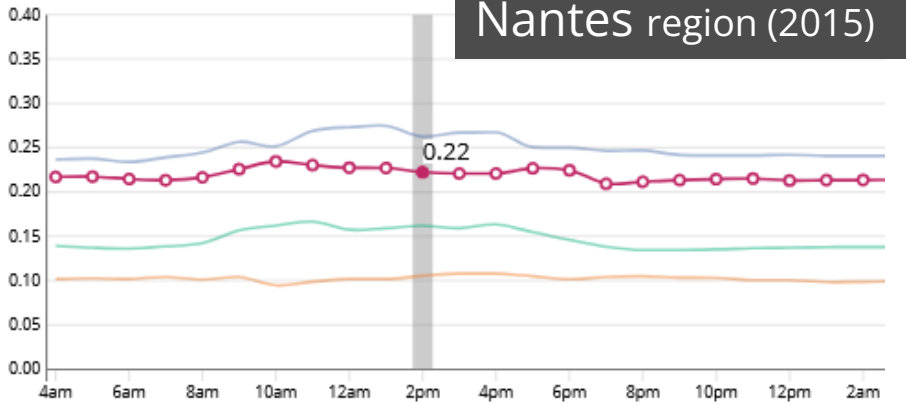
- Very high
- High
- Middle
- Low

DUNCAN VALUES IN 5 CITY REGIONS FOR **VERY HIGH EDUC.** SUBPGROUP

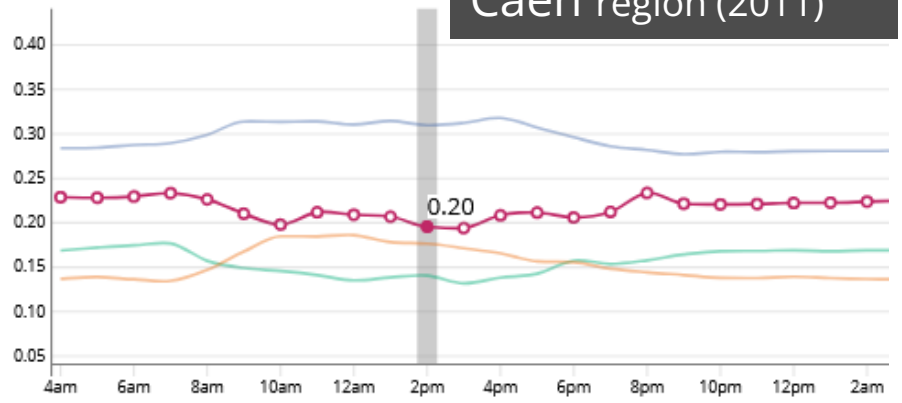
Bogotá region (2012)



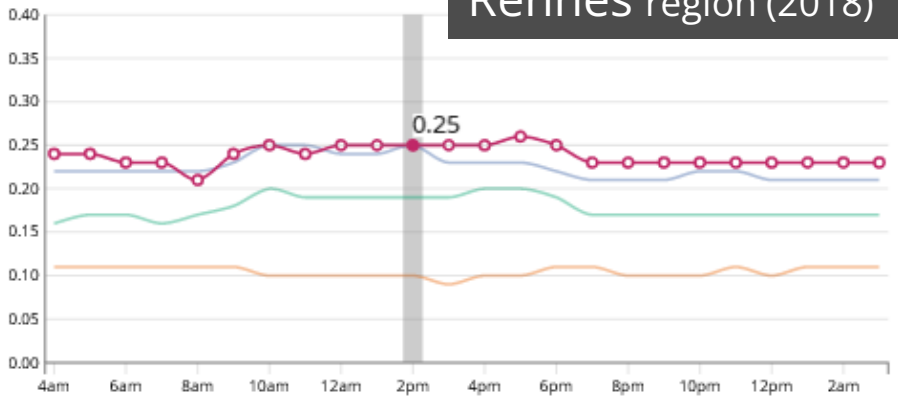
Nantes region (2015)



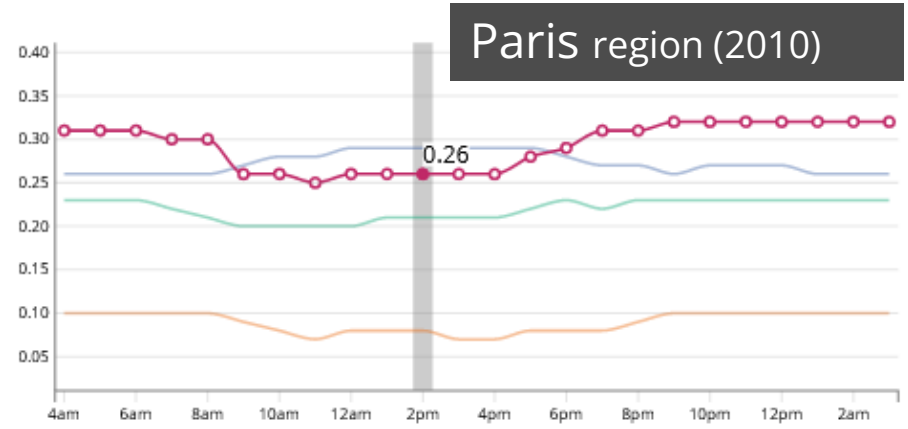
Caen region (2011)



Rennes region (2018)



Paris region (2010)



Methodological challenges for comparing cities

- Often risky to compare **crude** segregation values across cities
  - Due to differences in geographical boundaries (similar city definitions?) and intra-urban units (similar neighborhoods?)
  - & differences in population distributions for the outcome of interest (e.g., education)

- Less risky to examine **relative** segregation values across cities by comparing:
  - Over time (e.g., at 2 AM vs. at 2 PM)
  - Between subgroups (e.g., highly educated subgroup vs. low-educated subgroups)

SEGREGATION FOR GENDER & EDUCATION SUBGROUPS

IS SEGREGATION HIGHER AT NIGHT THAN DURING THE DAY ?

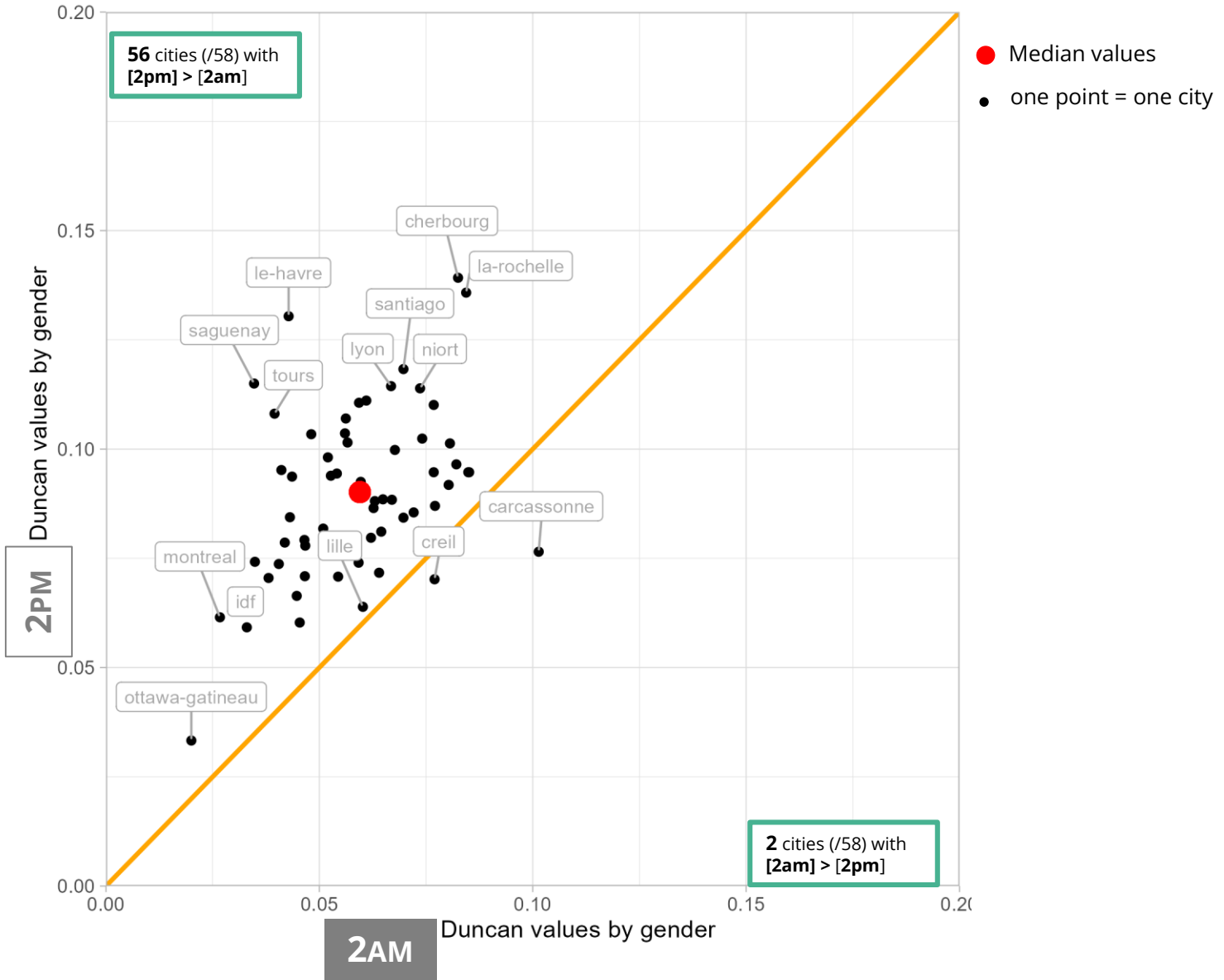
COMPARISON OF DUNCAN VALUES (2 AM *VS.* 2 PM) TO SEE IF DAILY MOBILITY REALLY MIXES SOCIAL GROUPS

QUESTION #1

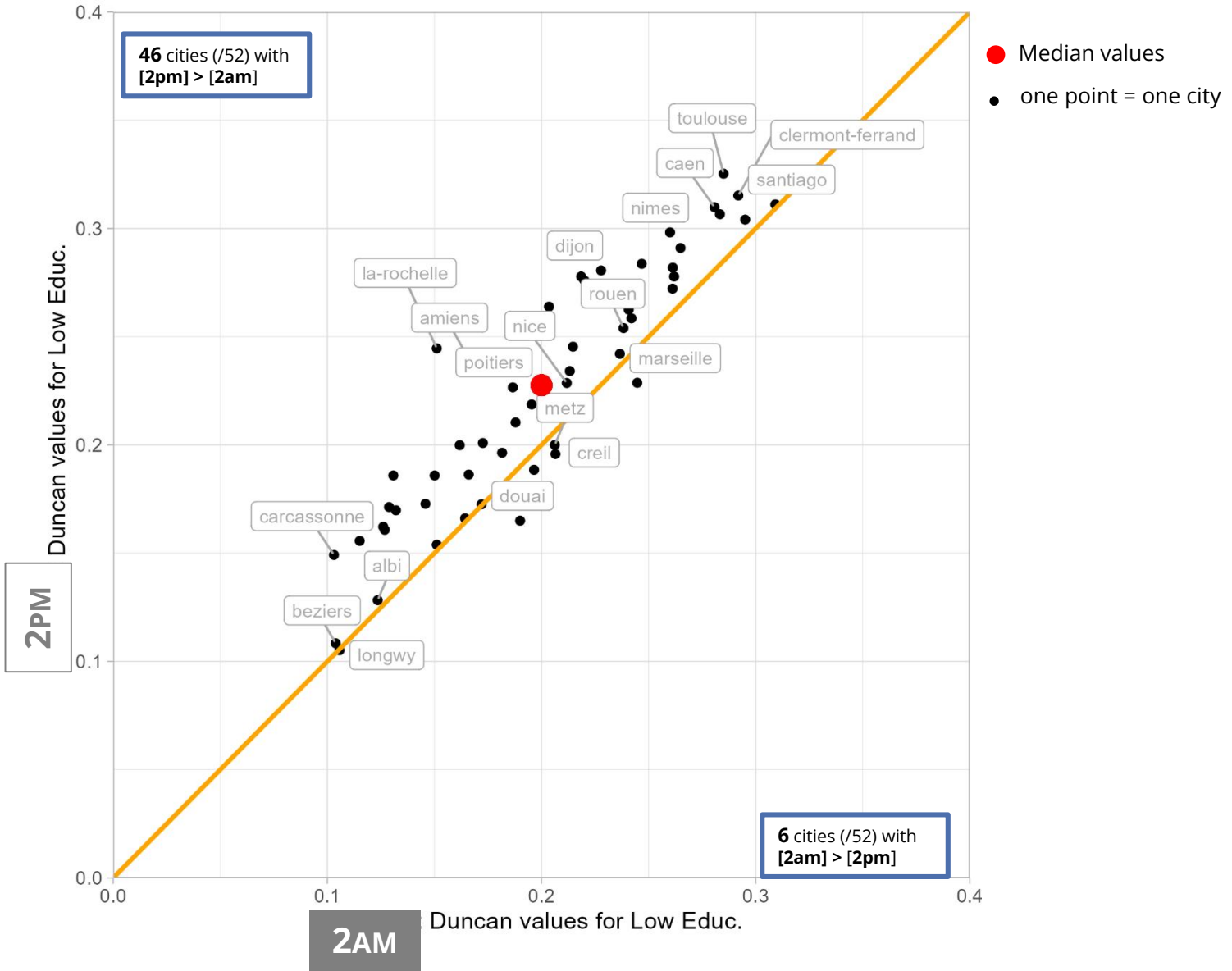
- ✓ ***Scientific*** *implications when exploring opportunities for social interaction, measuring social inequalities in proximity to services…*
- ✓ ***Political*** *implications related to daily transport, service locations, area-based interventions…*



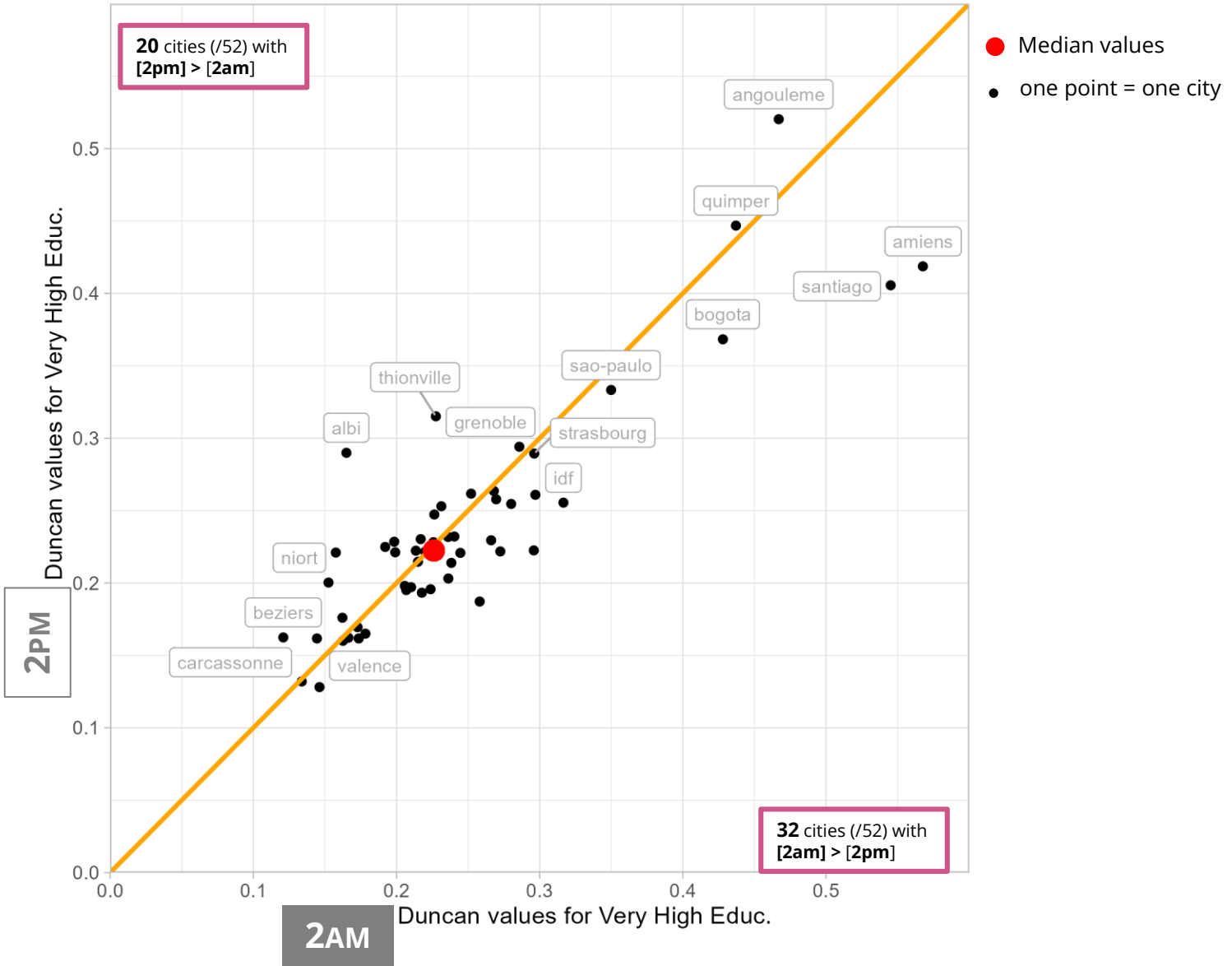
SEGREGATION **WOMEN** (MEN) AT 2AM *VS.* AT 2PM ACROSS 58 CITY REGIONS



SEGREGATION FOR **LOW-EDUCATED** PEOPLE AT 2AM *VS.* AT 2PM ACROSS 52 CITY REGIONS

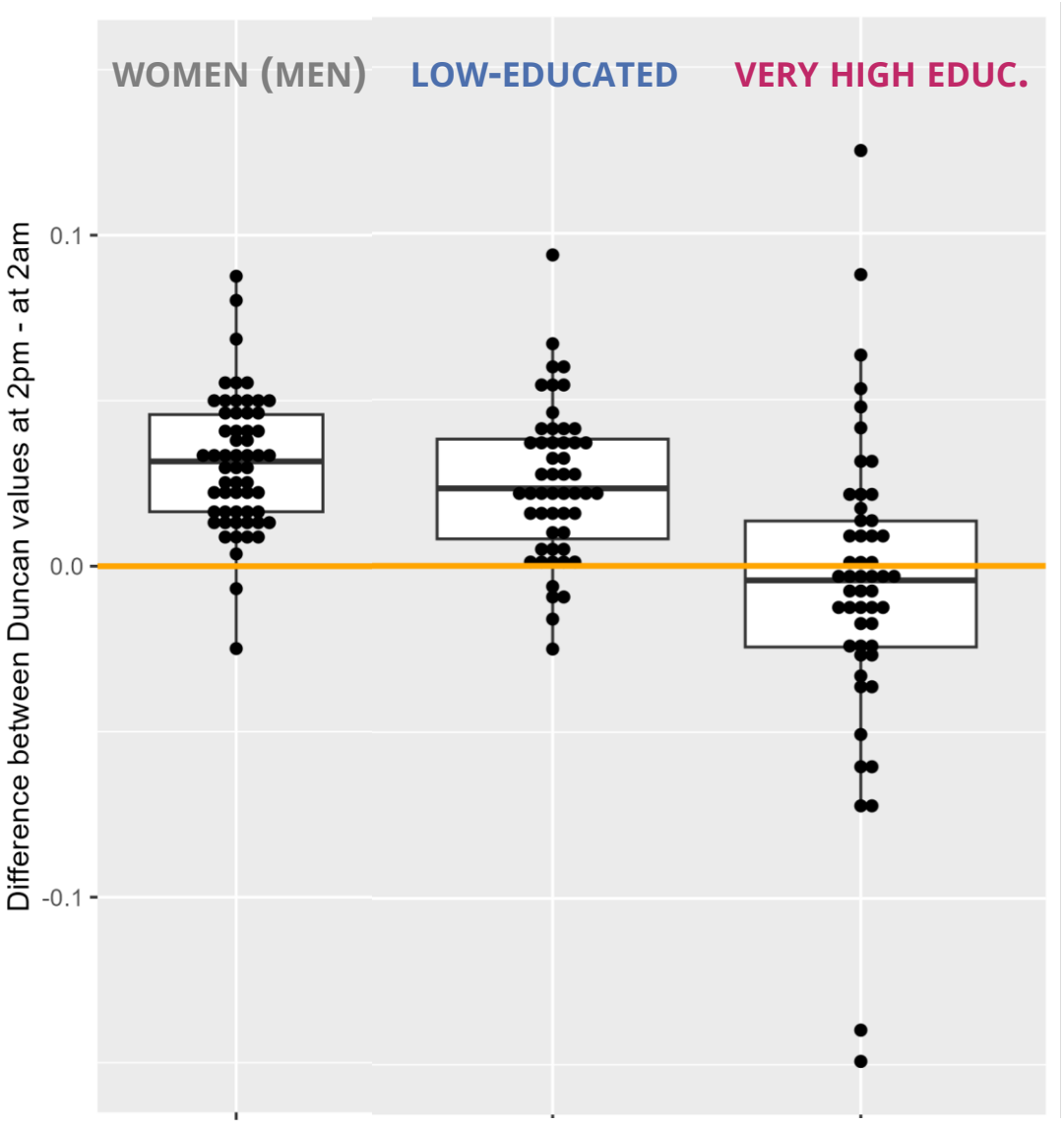


SEGREGATION FOR **VERY HIGH EDUCATED** PEOPLE AT 2AM *VS.* AT 2PM ACROSS 52 CITY REGIONS



DIFFERENCES IN DUNCAN VALUES BETWEEN 2PM - 2AM

ACROSS CITY REGIONS



- one point = one city

KEY FINDINGS - QUESTION 1

Is segregation higher during the day (2pm) than at night (2am)?

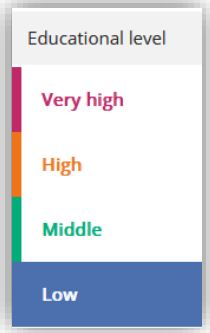
- Duncan values are found to be higher during the day than at night in a very large majority of cities for :
  - Gender
  - Low educated people

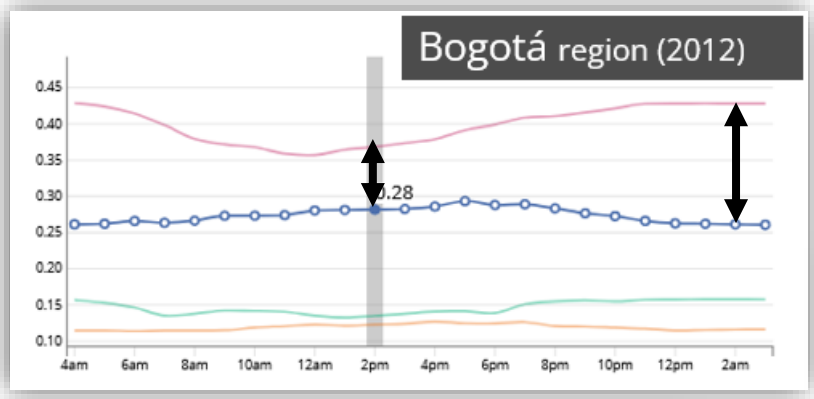
- But, various patterns across cities for:
  - Very high educated people

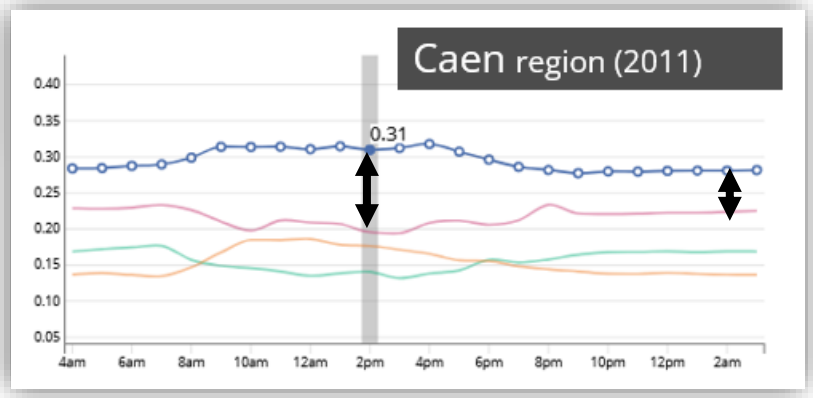
Question #2

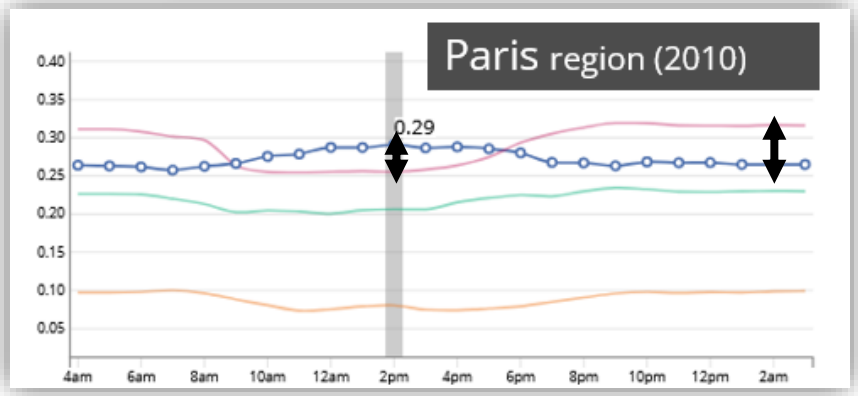
Which subgroup is the most segregated during the day ? The same than at night ?

Comparison of Duncan values (low-educated *vs.* very high-educated) at 2 pm and at 2 am









URBAN GEOGRAPHY
2023, VOL. 44, NO. 1, 1–36
https://doi.org/10.1080/02723638.2021.1959778

Routledge
Taylor & Francis Group

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Socio-economic segregation in European cities. A comparative study of Brussels, Copenhagen, Amsterdam, Oslo and Stockholm

Karen Haandrikman[a], Rafael Costa[b,c], Bo Malmberg[a], Adrian Farner Rogne[d] and Bart Sleutjes[c*]

ABSTRACT

The purpose of this study is to compare socioeconomic segregation patterns and levels in Brussels, Copenhagen, Amsterdam, Oslo, and Stockholm with uniform measurements. Previous research has been hampered by conceptual and methodological shortcomings. We use harmonized datasets containing geocoded indicators based on a nearest-neighbors approach. Our analyses offer an unprecedented comparison of patterns and levels of socio-spatial inequalities in European capitals at multiple scales. Using maps, segregation indices and percentile plots, we find that for all cities, the segregation of the rich is much stronger than the segregation of the poor. Macro-scale poverty segregation is most prominent in Stockholm and Brussels, and quite low in Amsterdam, while macro-scale affluence segregation is most pronounced in Oslo. At micro-scales, Brussels and Stockholm stand out with very high local poverty concentrations, indicating high levels of polarization. We interpret differences in the light of spatial inequalities, welfare regimes, housing systems, migration and area-based interventions.

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tion is most pronounced in Oslo. At micro-scales, Brussels and
concentrations,
differences in the
systems, migra-

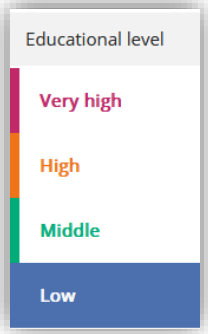
Findings from a residential-based approach.

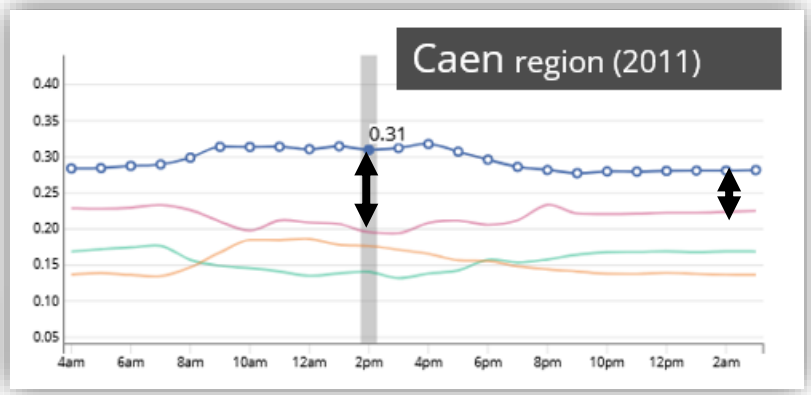
But is this still true when considering people's everyday mobility?

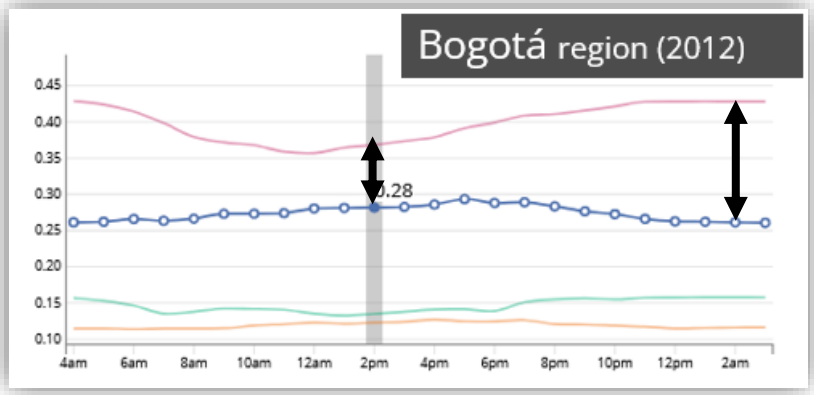
Let's compute the **differences** for each city :

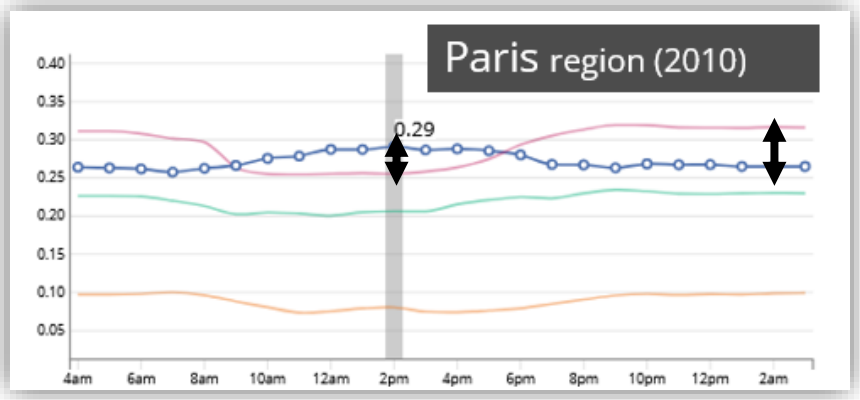
*Duncan[**Very high** educated subgroup] - Duncan[**Low** Educated subgroup]*

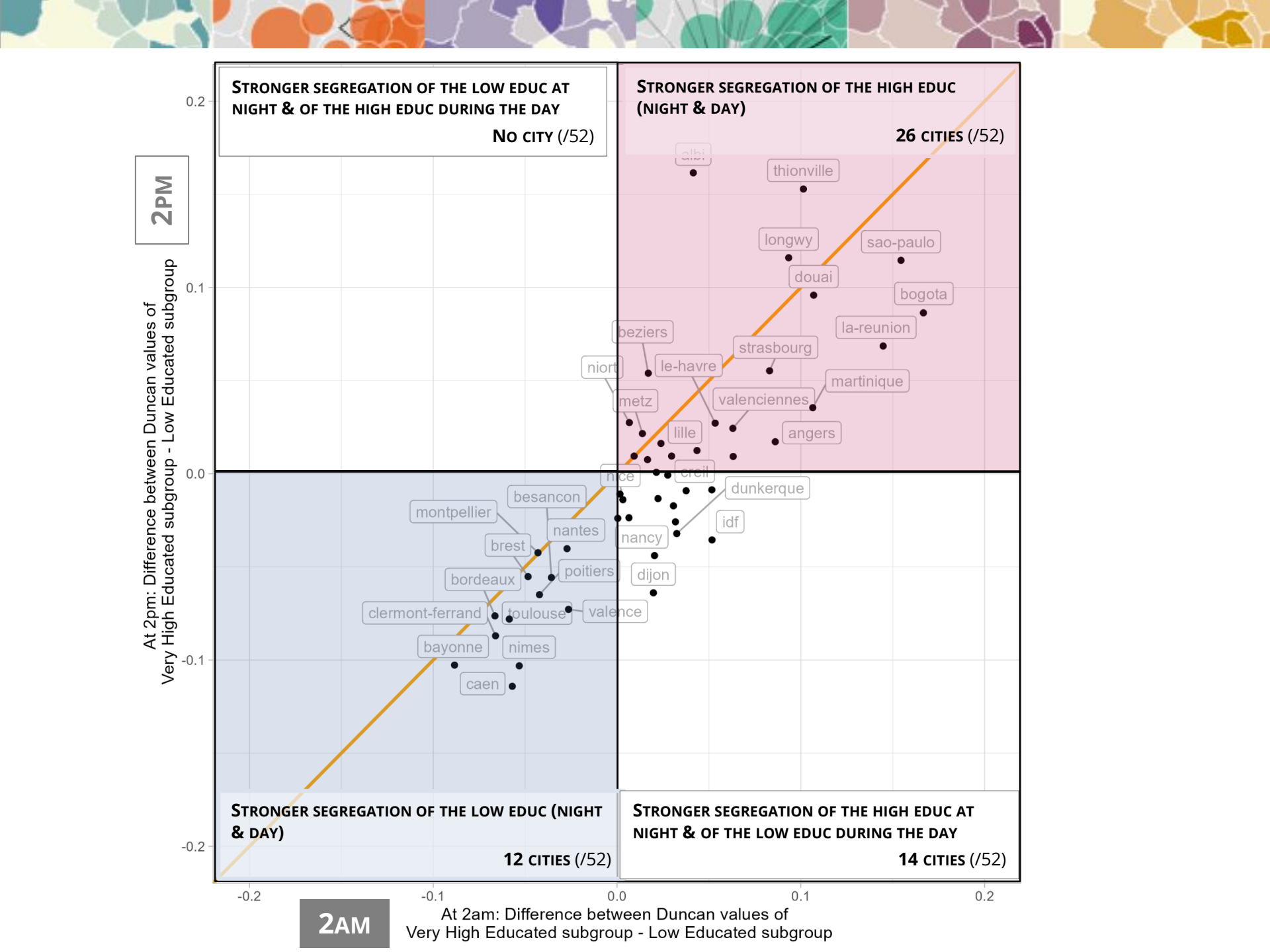
At **2AM** and At **2PM**










STRONGER SEGREGATION OF THE LOW EDUC AT NIGHT & OF THE HIGH EDUC DURING THE DAY
NO CITY (/52)
STRONGER SEGREGATION OF THE HIGH EDUC (NIGHT & DAY)
26 CITIES (/52)
STRONGER SEGREGATION OF THE LOW EDUC (NIGHT & DAY)
12 CITIES (/52)
STRONGER SEGREGATION OF THE HIGH EDUC AT NIGHT & OF THE LOW EDUC DURING THE DAY
14 CITIES (/52)
2PM
2AM
At 2pm: Difference between Duncan values of Very High Educated subgroup - Low Educated subgroup
At 2am: Difference between Duncan values of Very High Educated subgroup - Low Educated subgroup
0.2
0.1
0.0
-0.1
-0.2
-0.2
-0.1
0.0
0.1
0.2
albi
thionville
longwy
sao-paulo
douai
bogota
la-reunion
beziers
strasbourg
niort
le-havre
martinique
metz
valenciennes
lille
angers
nice
creil
dunkerque
besancon
montpellier
idf
nantes
nancy
brest
poitiers
dijon
bordeaux
clermont-ferrand
toulouse
valence
bayonne
nimes
caen

KEY FINDINGS - QUESTION 2

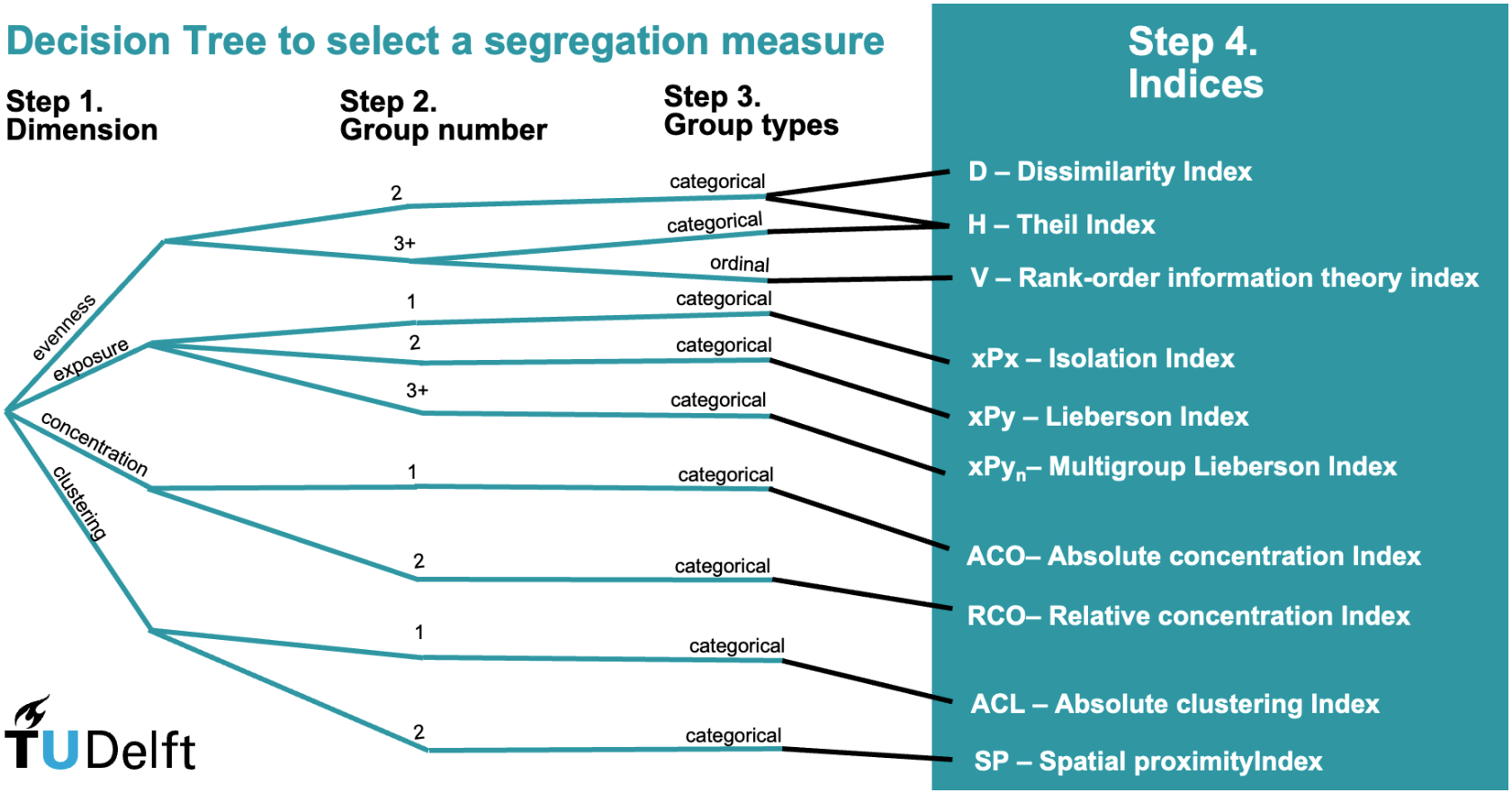
- Empirical proof of the need not to draw conclusions about the most segregated social groups from data concerning
  - a single city
  - a single time of day

- Go further to relate these hourly segregation patterns to the characteristics of the cities (size, organisation, country, etc.).

IV. Perspectives

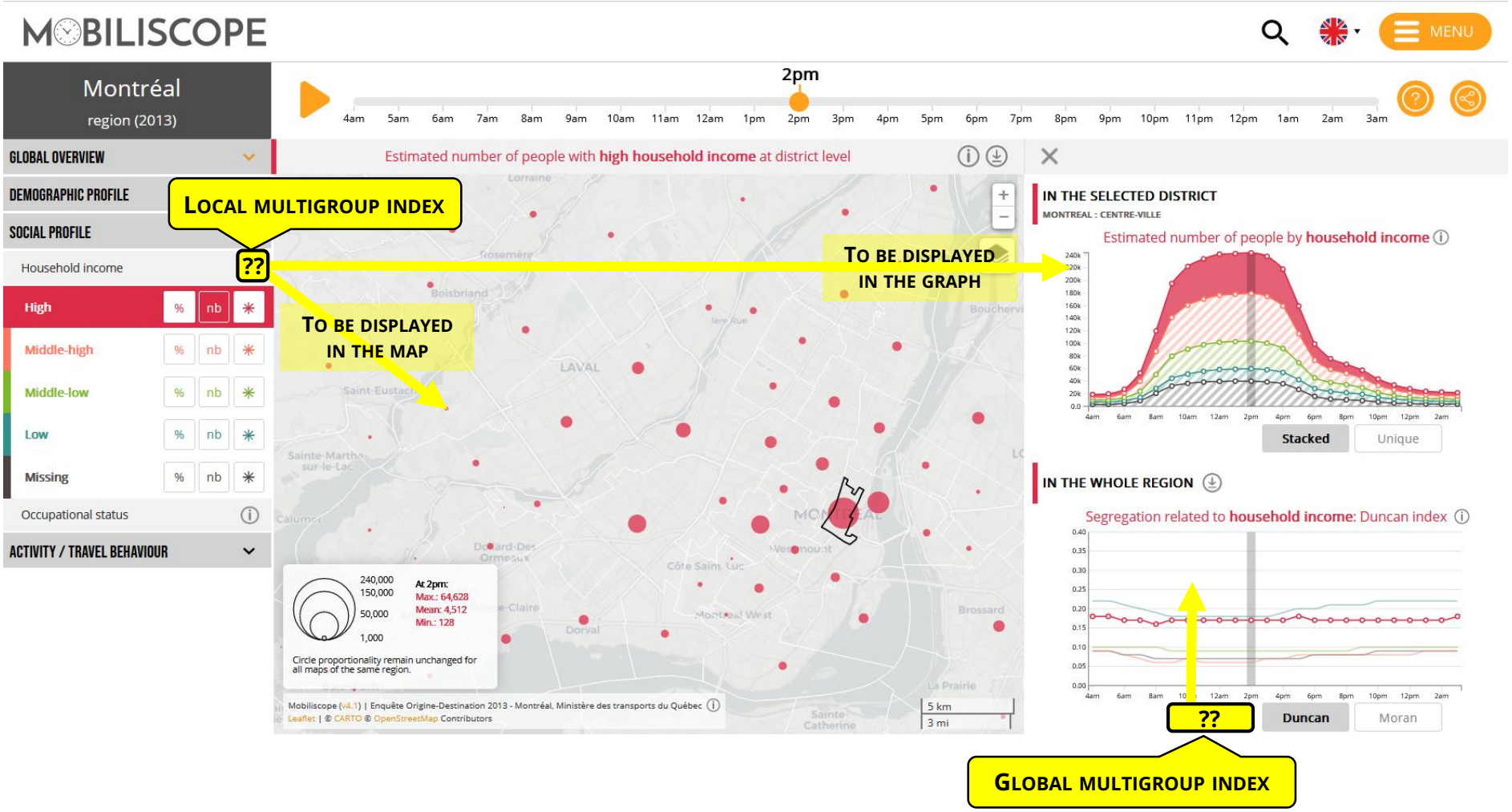
1. Add new measures of segregation in the Mobiliscope ?

ADD NEW MULTIGROUP MEASURES OF SEGREGATION ?



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ADD NEW MULTIGROUP MEASURES OF SEGREGATION ?



IV. Perspectives

1. Add new measures of segregation in the Mobiliscope ?

2. Explore hourly segregation across years from next Mobiliscope version

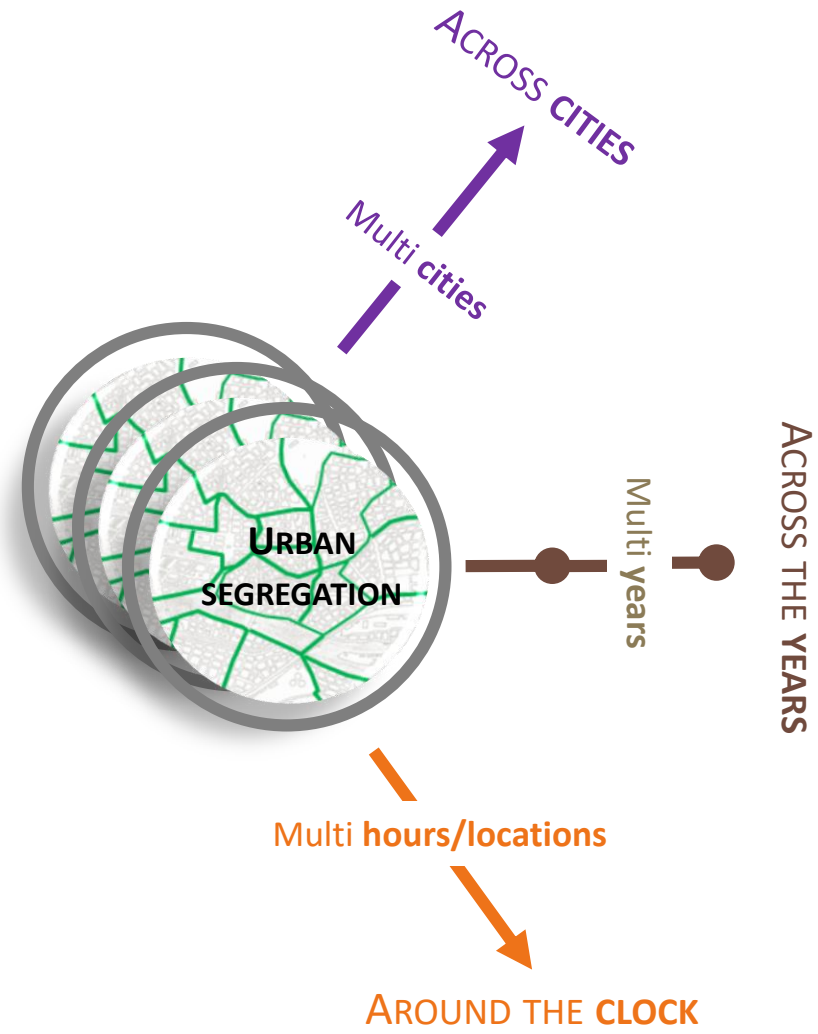
Next Mobiliscope version: multi-dates data for several French cities



!!! Comparability challenges over the years :

- ≠ City boundaries
- ≠ Neighborhoods boundaries

⟶ Creation of a ‘multi-date’ geographical zoning



TO BE ABLE TO EXPLORE HOURLY VARIATIONS IN SEGREGATION ACROSS CITIES … *AND ACROSS THE YEARS*



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THANKS FOR YOUR ATTENTION

TIME FOR EXCHANGE AND DISCUSSION

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